Interactive Trading Simulator

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Abstract

The “Interactive Trading Simulator” is a comprehensive web-based educational platform designed to help beginners learn stock trading in a risk-free environment. Using Python’s Streamlit framework, the application enables users to practice executing trades, analyzing market data, and understanding fundamental trading concepts. The simulator incorporates historical stock data, interactive charts, portfolio tracking, and a structured educational guide. This report describes the methodology, technical implementation, development process, challenges encountered, results, and self-evaluation, highlighting the effort and learning outcomes associated with the project.

Keywords: trading simulator, stock trading education, interactive learning, portfolio management

Project Description

The “Interactive Trading Simulator” is designed to bridge the gap between theoretical knowledge and practical application in stock trading, providing an educational platform that transforms abstract concepts into hands-on experiences. Many beginners are intimated by financial markets due to the inherent risk of monetary loss, which can discourage learning and experimentation. By offering a controlled risk-free environment, this platform enables users to make informed trading decisions, observe the consequences of their decisions on portfolio performance, and gain insights from mistakes without facing financial repercussions. This experiential learning approach allows users to develop confidence, refine strategies, and build a foundational understanding of market mechanics in a safe and supportive setting.

The platform is structured around four main components that guide users through a logical progression from introduction to practical simulation The “Start Here” module serves as an onboarding experience, providing an overview of the platform’s structure, functionalities, and navigation tools. It ensures that users understand how to load stock, navigate between pages, and begin engaging with the simulator effectively. The “Beginner’s Guide” component acts as a repository of educational resources, covering trading fundamentals, order types, risk management strategies, and common pitfalls, equipping users with the knowledge approach trading scenarios confidently.

At the heart of the platform lies the “Trading Simulator”, which allows users to apply theoretical knowledge using historical stock data. Users can execute buy and sell orders, track real-time portfolio performance, and analyze detailed trade histories. This gives users the opportunity to review past decisions, identify patterns, and learn from both successes and mistakes. Finally, the “Support Center” offers direct assistance to users through a frequently asked questions section and a built-in contact form. This component reinforces the educational value of the platform by combining self-guided learning with accessible support, creating a comprehensive and engaging environment for developing trading skills. Together, these four components provide a structured, interactive, and user-friendly system that empowers beginners to transition from theoretical understanding to practical competence in stock trading.

Methodology

Architecture and Framework Selection

The project utilizes Streamlit as a primary development framework due to its ease of prototyping and intuitive interface for data-driven applications. Streamlit's session state allows persistent data, which is essential for maintaining portfolio states, trade histories, and historical navigation. The application follows a modular design with separate files for each major components – start_here.py, beginners_guide.py, trading_simulator.py, and support_center.py – coordinated through dashboard.py for multi-page navigation. This structure ensures maintainability and allows independent development and testing.

Data Processing and Visualization

Historical stock data is loaded from CSV files (containing date, open, high, low, close, adjusted close, and volume information) using *pandas*. The workflow automatically parses dates and sorts the data chronologically to ensure accurate simulation. Real-time price calculations use the current simulated day, while portfolio tracking updates cash balances, position values, and profit/loss metrics dynamically. Trade execution supports both market and limit orders, including commission fees and slippage for realistic conditions. For visualization, the platform employs *plotly* to generate interactive candlestick charts, with trade markers overlaid to indicate buy orders (green upward triangles) and sell orders (red downward triangles), providing users with immediate visual feedback on their trading decisions.

Session State Management

Persistent session state tracks the current simulated day, cash balance ($100,000 starting), share positions, trade history (with timestamps, prices, and commissions), and loaded stock data/metadata. A reset feature with confirmation prevents accidental data loss.

Educational Content Integration

The "Beginner's Guide" uses expandable sections to explain trading concepts like order types, risk management, and common pitfalls. External resources include educational websites and YouTube channels.

Development Process

Milestone/Week 1: Data Collection and Dashboard Planning

Jiang focused on collecting historical stock data and preprocessing it to ensure it was clean and properly formatted. She also spent time brainstorming and designing the layout of the dashboard to create an intuitive user experience. This included planning the overall structure, deciding on page navigation, and conceptualizing an intuitive user experience that would allow users to explore trading simulations, track portfolios, and access educational content seamlessly.

Milestone/Week 2: Trading Simulator Development

Jiang implemented the core features of the "Trading Simulator", including CSV data loading with error handling. She added a day-by-day navigation system (±1, ±10 days) and developed trade execution logic with validation. Interactive charts with trade markers were created, and portfolio value and profit/loss tracking were integrated.

Milestone/Week 3: Educational Content and Support Integration

Jiang worked on integrating the educational and support content, adding the "Start Here" page, the "Beginner's Guide" with expandable trading lessons, and the "Support Center" for user assistance. This week also included designing the content layout and ensuring visual consistency.

Milestone/Week 4: Testing, Analysis, and Refinement

Jiang conducted thorough testing of the application, analyzed the results to identify areas for improvement, and applied final refinements to ensure a smooth, visually consistent, and polished user experience. Any bugs or inconsistencies discovered were addressed, and the interface was refined to enhance usability.

Milestone/Week 5: Demo Preparation and Presentation

Jiang prepared a comprehensive demo showcasing the functionality, usability, and educational features of the platform, and delivered a presentation highlighting the project's capabilities. She organized the presentation to clearly communicate the platform's purpose, design decisions, and key features, ensuring that both technical and non-technical audiences could understand and appreciate the project.

Challenges and Solutions

During the development of the platform, Jiang encountered several challenges that required careful solutions. Rendering full historical data along with all trade markers caused performance issues, so she optimized the charts by limiting the visible data and selectively displaying trade markers, resulting in a smoother and more responsive experience. Implementing accurate trade execution for both market and limit orders, while accounting for validation, commissions, and error handling, was complex; Jiang addressed this by separating the logic into dedicated functions with thorough checks, ensuring realistic and reliable trading behavior. Additionally, supporting both preset datasets and user-uploaded CSV files while maintaining consistent portfolio states was challenging. To solve this, she developed a dual loading system with an automatic portfolio reset, preserving data integrity and preventing errors during uploads.

Results and Analysis

To evaluate the platform from a beginner's perspective, Jiang asked a fellow acquaintance, Jenny Huynh, who has no prior knowledge of finance or trading, to test the simulator. Huynh's feedback provides insight into the platform's usability, educational value, and technical performance. All planned features were successfully implemented, and the platform allowed users to "move through different stocks and see everything update instantly," creating an experience that "felt complete and easy to use" (Huynh). The interface guided users through onboarding, educational content, simulation, and support with step-by-step navigation. Huynh appreciated that someone who "didn't know anything about trading" could navigate without feeling overwhelmed. She suggested using green for price increases and red for decreases (referencing candlestick charts instead of triangles for better visibility), which would make it easier for beginners to quickly interpret market movements and enhance the overall clarity of the simulator. The simulator effectively teaches beginners through practical examples, immediate feedback, clear visuals of portfolio impact, and a risk-free environment. Users can "learn how buying and selling affects my portfolio" and "try things without worrying about real money" (Huynh).

Limitation and Future Work

The Interactive Trading Simulator has some limitations. It only uses historical stock data, so users cannot experience real-time market changes. Advanced trading strategies, like options or margin trading, are not included. The simulator also supports only one portfolio at a time, which limits experimenting with multiple strategies. In the future, real-time market data, advanced trading options, multiple portfolios, and mobile support could be added to make the simulator more realistic and flexible for learners.

Effort Justification

This project required much more effort than a typical assignment, meeting the 1.5x effort requirement. Building the “Interactive Trading Simulator” involved creating a multi-page Streamlit app, managing session state, making interactive charts with trade markers, and implementing realistic trade execution with fees and slippage. Educational content was also developed for beginners, including guides and onboarding instructions. Throughout the project, new skills were learned, such as session state management, data visualization, trade logic, and user interface design. These combined tasks made the project more complex and time-consuming than standard coursework.

Rubric Criteria

The project meets all rubric criteria in several areas. Core features work as planned, including multi-page navigation, CSV data loading, day-by-day stock navigation, trade execution, portfolio tracking, and trade history display. The user interface is clear and easy to use, with interactive charts and helpful error messages. Educational content is complete, with guides, curated resources, and onboarding instructions. The code is well-organized, documented, and efficient, with strong error handling. Extra features go beyond the basic requirements, such as market and limit orders, configurable chart views, trade markers, portfolio reset, and a support center with email integration. Overall, each category meets or exceeds expectations, showing the project’s high quality and scope.

Conclusion

The “Interactive Trading Simulator” successfully combines educational content with practical trading simulations. Beginners can navigate the platform, execute trades, and understand market mechanics without financial risk. The project demonstrates substantial technical and educational effort, provides immediate feedback for learning, and equips users with foundational trading knowledge. With further enhancements, the platform could expand to more advanced trading features and real-time data.